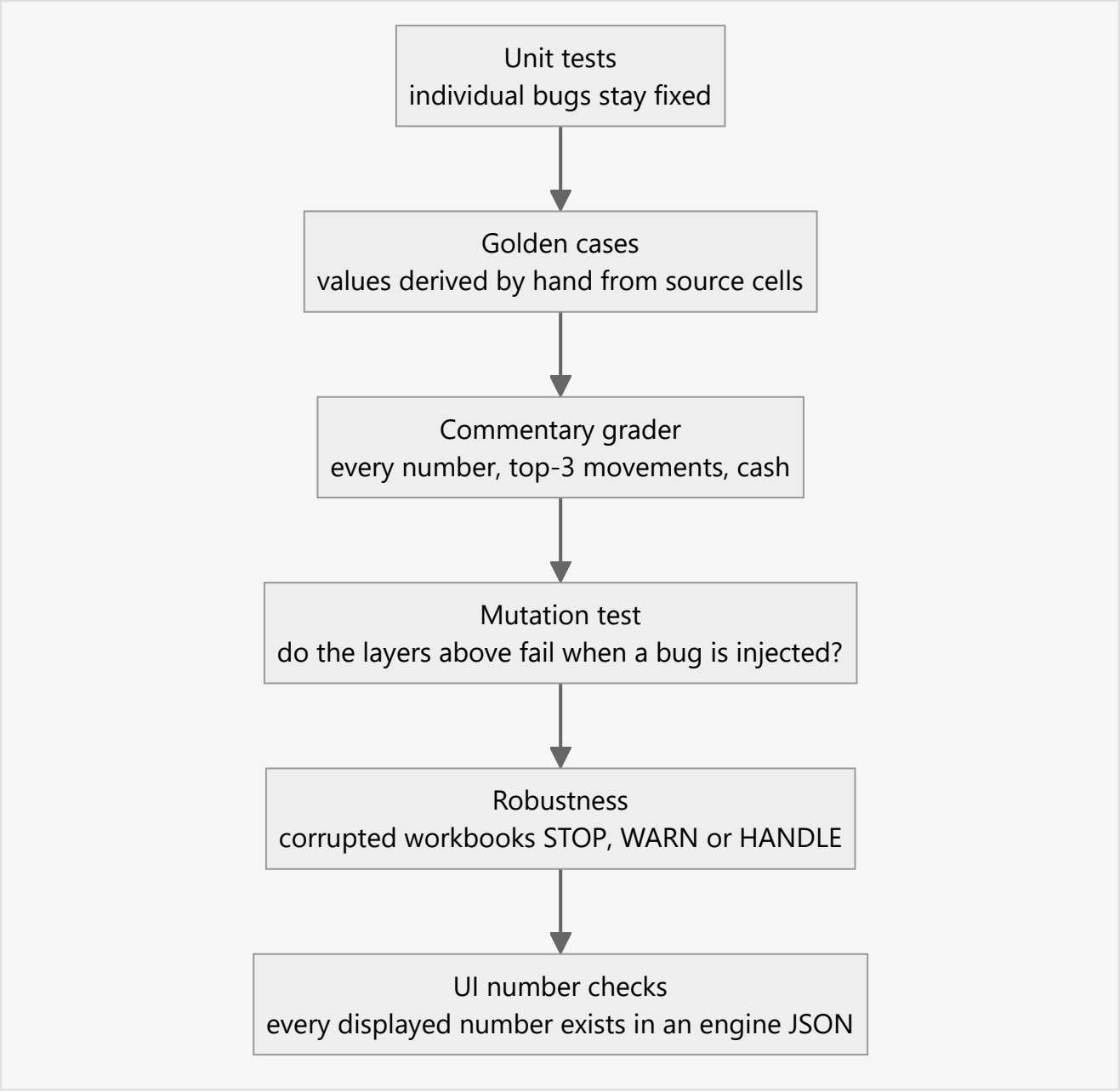
Evals and Regression

How we show that the numbers and the commentary are right, and that the checks themselves can fail.

Eval layers



Layer	Command	What it proves	Latest
Unit tests	`python -m unittest discover -s tests`	Bugs stay fixed; client isolation; redaction; workflow failure messages	129 run, 129 OK
Golden cases	`python 03_evals/run_golden_evals.py`	Engine numbers and flags equal values derived by hand	22/22

Layer	Command	What it proves	Latest
Commentary grader	`python 03_evals/grade_commentary.py`	Every number matches the engine JSON; top-3 movements named; cash and runway covered	Engine, reviewed and saved-skill text PASS; flawed control FAILs as intended
Mutation test	`python 03_evals/mutation_test.py`	The tests fail when a known bug is injected	13/13 caught
Robustness	`python 03_evals/robustness/fault_injection.py`	Corrupted workbooks behave as declared on both entry points	32/32 pass, source workbook hash unchanged
UI number checks	`python -m unittest tests.test_ui_numbers`	No displayed number is computed in the UI	15 tests OK

Golden case table

22 cases in three groups. "Real workbook" means the ABC workbook read in memory, never modified.

Source	Count	IDs	What the group is
`workbook`	4	G01, G02, G03, G06	Real data already triggers the rule; no cell changed
`workbook+perturbation`	9	G04, G05, G07, G08, G14, G19–G22	One or two cells changed to trigger a rule; G19–G22 require the intake gate to refuse a perturbed copy
ABC cell-sourced	9	G09–G13, G15–G18	Expected values computed by hand from cached source cells, not by the engine

Case	Pins
G01–G03, G06	MoM revenue, gross margin, TTM needs twelve consecutive months, expense category trend
G04, G05, G07, G08, G14	Duplicate month rejected; missing revenue rejected; zero prior revenue gives a null %; short TTM disclosed; month labels survive normalization
G09–G10	February is the peak marketing month; the 5 one-off items, Dec 2025 the event month
G11–G13	First valid TTM month is Dec 2025; largest OPEX spikes; full June 2026 summary
G15–G17	Platform Revenue step-down in Jan 2026; Bad Debts reversal in Jun 2026; cash 713,481.26 and runway 3.40 months
G18	Gross margin falls over 5 pp in Mar 2025, Aug 2025 and Apr 2026 only
G19–G22	Intake STOP by family: structure, values, footing, coverage

Tolerances: currency ±0.005; ratios, percentages and margins ±0.000001; identity fields exact. Below 100% fails.

Mutation results

The runner refuses to start unless the unmodified copy is green. A mutant is an eval gap if nothing fails.

Mutation	Files mutated	Caught
Reverse month order	`financial_analysis.py`, `abc_workbook_adapter.py`	yes, yes
Flip revenue sign	both	yes, yes

Mutation	Files mutated	Caught
Sum subtotals twice	`abc_workbook_adapter.py`	yes
Shift months by one	both	yes, yes
Drop one opex category	both	yes, yes
TTM over 11 months	both	yes, yes
Wrong margin denominator	both	yes, yes

Total: 13/13 caught (7 bug classes). The wrong-denominator mutant is caught by G02, G13, G18 and two unit tests.

Robustness matrix summary

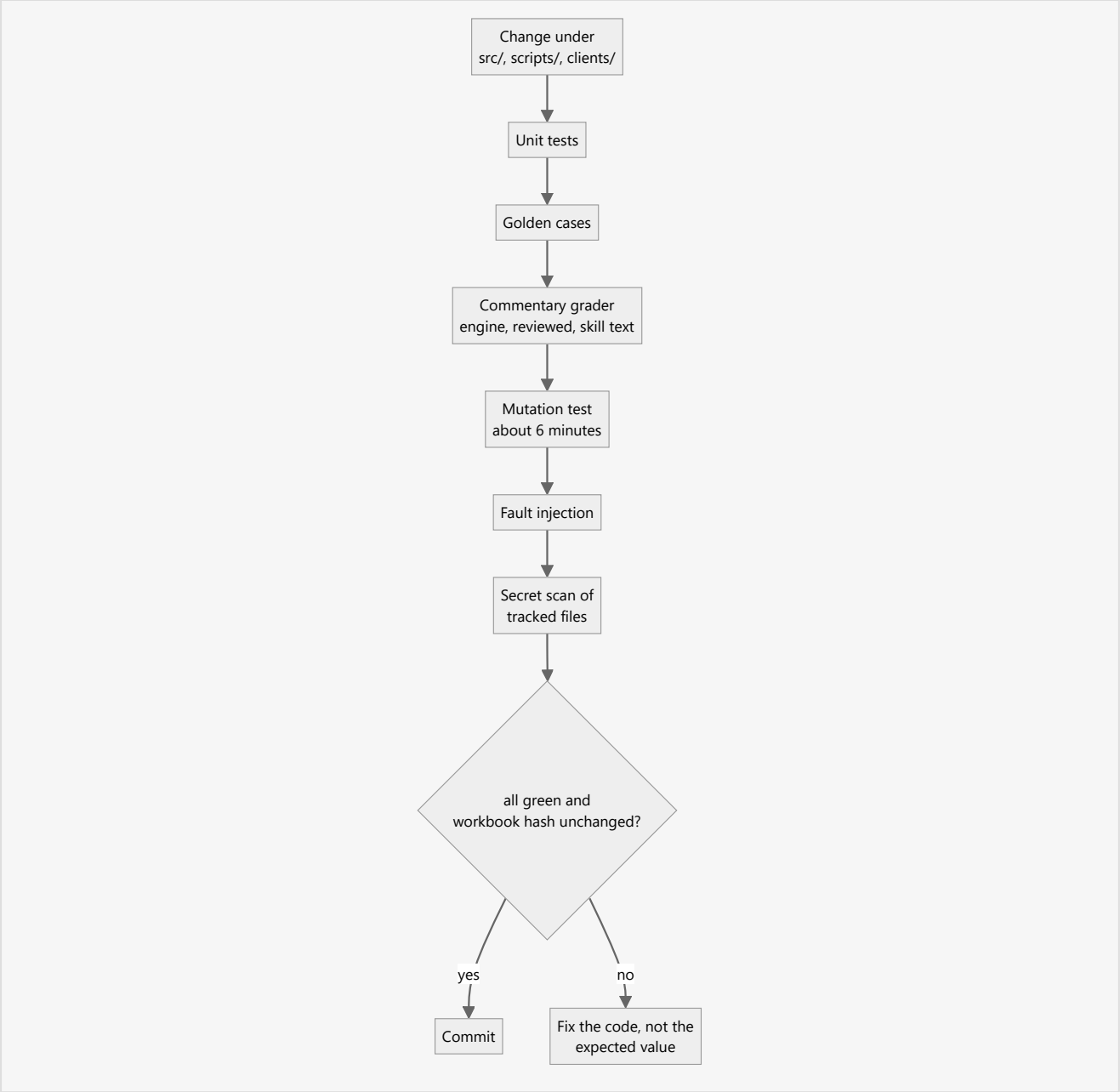
Each fault is applied to a copy of the ABC workbook and run through both entry points that produce numbers (`run_pipeline.py` and step 1 of the skill). The required behaviour is fixed before the run. **STOP** means intake refuses with a message naming the cell or sheet and writes no packet. **WARN** means the run continues and the warning reaches the packet and the Data Quality sheet. **HANDLE** means no new warning and every number equals the baseline. **SILENT-WRONG** (success, different numbers, no warning) must never occur.

Family	Faults	STOP	WARN	HANDLE	Examples
Structure	11	6	1	4	Prior month deleted; duplicated month; sheet renamed; merged headers
Values	9	4	3	2	Text in a numeric cell; Excel `#DIV/0!`; x1000 outlier; zero revenue; infinity
Labels	7	2	3	2	Renamed account; look-alike character; deleted section total
Coverage	5	4	1	0	Only 11 months; formulas without saved values; corrupt or locked file
Total	**32**	**16**	**8**	**8**	

Result: 32/32 pass, source workbook hash unchanged. Full table with the message shown for each fault is in the Verification and Robustness Report.

Regression procedure

A change is blocked by any of: a golden case that passed now fails; any unit test fails; a commentary number mismatch, a dropped top movement or lost cash coverage; a flag that changes behaviour; the source workbook's bytes changing; a caught mutant that now survives; a client-isolation failure; a new data-quality warning that no one reviewed; any fault that stops behaving as declared, or any SILENT-WRONG.



Expected values in the evals are never changed to make a test pass. If one looks wrong, the source-cell evidence (sheet, cell, value or formula) is shown and the owner decides. Wording changes that keep every number and movement are not regressions.